

CSTM: Canonical Space Transformation

Three cameras, same scene, different focal lengths

Ultra-wide



$f = 14\text{mm}$ 4032×3024
 $\omega = 1.71$



Enlarged $1.71 \times$

Wide (canonical)



$f = f^c = 24\text{mm}$ 5712×4284
 $\omega = 1.0$

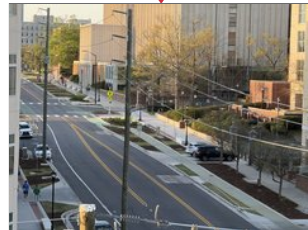


Unchanged

Telephoto



$f = 120\text{mm}$ 4032×3024
 $\omega = 0.2$



Shrunk $0.2 \times$

CSTM-image: $I_c = \text{resize}(I, \omega)$ where $\omega = f^c / f$